Reduced Optimal Control with PDE Constraints: Model, Discretization, and Optimization Methods

Fachseminar

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Abstract

We study a linear-quadratic optimal control problem constrained by the Poisson equation with homogeneous Dirichlet boundary conditions. The control acts as a volumetric source on a subdomain $\omega \subset \Omega$. Using P1 finite elements on a family of pre-generated meshes, we eliminate the state and obtain a reduced functional $F(u)$ whose U-gradient is computed via an adjoint solve and the control-space mass matrix.

For each mesh, we solve the reduced problem in two ways: (i) by Conjugate Gradients (CG) on the SPD stationarity system $Q\mathbf{u} = \mathbf{b}$, and (ii) by first-order reduced-space methods (Barzilai–Borwein, fixed-step gradient descent with step size $1/L$, and constant-parameter Nesterov based on (m, L)). We report convergence histories in $\|\nabla_U F(\mathbf{u}_k)\|_U$, final objective values, iteration counts, and timing components, and we perform a mesh-independence study over multiple mesh sizes.

The experiments show consistent final objective values across methods and stable objective/norm quantities under mesh refinement. On the representative mesh, first-order methods reach the stopping criterion in very few updates from the chosen initialization, while CG provides the corresponding stationarity-system reference solution.

1 Problem Statement

We consider a linear-quadratic optimal control problem governed by the Poisson equation on a polygonal domain $\Omega \subset \mathbb{R}^2$ (in computations, typically $[0,1]^2$). A circular subregion $\omega \subset \Omega$ with nonzero Lebesgue measure is tagged on the mesh and serves as the support of the control. The control acts as a volumetric source term restricted to ω:

$$-\Delta y = \chi_\omega \, u \quad \text{in } \Omega, \tag{1}$$

$$y = 0 \quad \text{on } \partial\Omega, \tag{2}$$

where χ_ω denotes the characteristic function of ω. We denote by $V = H_0^1(\Omega)$ the state space and by $U = L^2(\Omega)$ the control space. For a given target $y_d \in L^2(\Omega)$ and a regularization parameter $\beta > 0$, we minimize

$$J(y, u) = \tfrac{1}{2} \int_\Omega (y - y_d)^2 \, dx + \tfrac{\beta}{2} \int_\Omega u^2 \, dx. \tag{3}$$

The target is chosen as the pyramid-like profile

$$y_d(x) = 0.5 - \max\left(|x_1 - 0.5|, \, |x_2 - 0.5|\right), \tag{4}$$

which belongs to $L^2(\Omega)$. The control enters only on ω via χ_ω, so the effective source equals u on ω and vanishes on $\Omega \setminus \omega$. The PDE with homogeneous Dirichlet boundary conditions is well-posed for $u \in L^2(\Omega)$, yielding a unique weak solution $y \in V$ by Lax–Milgram.

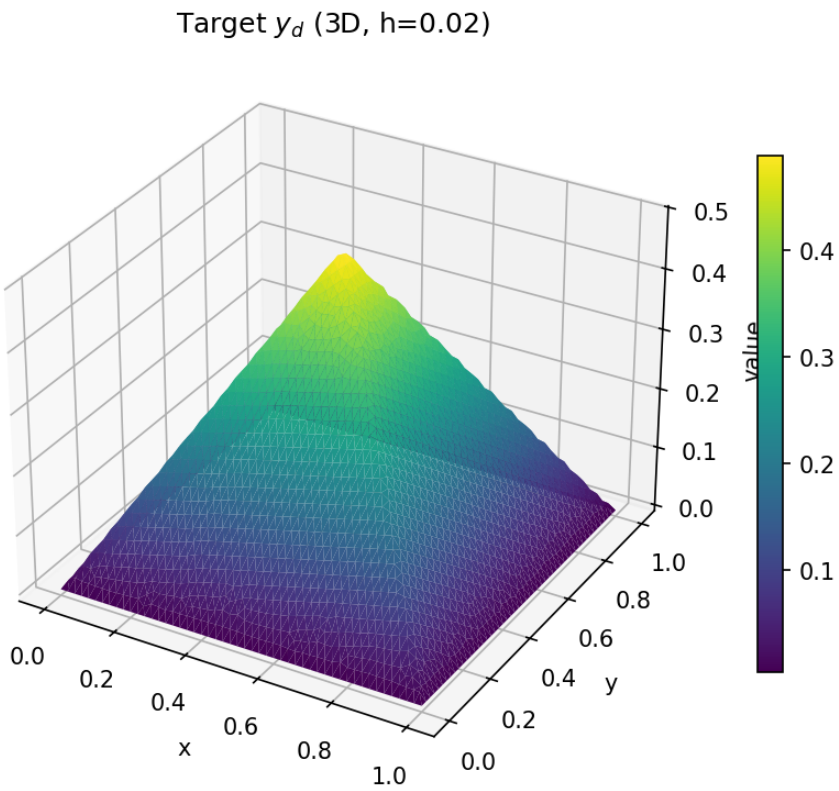


Figure 1: 3D surface of the target profile y_d on Ω.

2 Variational Formulation and Discretization

Weak formulation. Define the bilinear and linear forms

$$a(y,v) = \int_\Omega \nabla y \cdot \nabla v \, dx \quad \text{for } y, v \in V, \qquad b(u,v) = \int_\omega u \, v \, dx \quad \text{for } u \in U,\ v \in V. \tag{5}$$

For any $u \in L^2(\Omega)$, the weak state equation reads: find $y \in V$ such that

$$a(y,v) = b(u,v) \quad \forall v \in V. \tag{6}$$

By continuity and coercivity of $a(\cdot,\cdot)$ on V and boundedness of $b(\cdot,\cdot)$, this admits a unique $y \in V$.

Finite elements. Let $\mathcal{T}_h$ be a conforming triangulation of Ω. We use the conforming P1 space $V_h \subset V$ with nodal basis $\{\varphi_i\}_{i=1}^n$. The control is represented in the same P1 basis but measured with the L^2 inner product ("control metric"). We assemble the matrices

$$A_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \quad M_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx, \quad B_{ij} = \int_\omega \varphi_j \, \varphi_i \, dx, \quad (M_U)_{ij} = \int_\Omega \varphi_j \, \varphi_i \, dx. \tag{7}$$

Here, A is the stiffness matrix for state/adjoint solves, M is the state mass matrix used to weight the tracking term, B maps control coefficients to the state RHS via restriction to ω, and M_U is the control-space mass matrix defining the L^2 inner product in U_h.

Boundary conditions. Homogeneous Dirichlet boundary conditions are imposed on the state/adjoint by modifying A and setting the RHS at boundary degrees of freedom to zero when solving. Importantly, no boundary conditions are applied to B or M_U; these operators remain the unmodified L^2-based forms on Ω.

Discrete state equation and objective. With coefficient vectors $\mathbf{y}, \mathbf{u} \in \mathbb{R}^n$ and target coefficients $\mathbf{y}_d \in \mathbb{R}^n$, the discrete state equation and cost read

$$A\mathbf{y} = B\mathbf{u}, \tag{8}$$

$$J(\mathbf{y}, \mathbf{u}) = \tfrac{1}{2}(\mathbf{y} - \mathbf{y}_d)^\top M(\mathbf{y} - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U \mathbf{u}. \tag{9}$$

3 Reduced Formulation

State elimination and reduced functional. The discrete PDE constraint $A\mathbf{y} = B\mathbf{u}$ defines the linear control-to-state map $\mathbf{y}(\mathbf{u}) = A^{-1}B\mathbf{u}$. Substituting into the discrete objective yields the reduced functional

$$F(\mathbf{u}) = \tfrac{1}{2}(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d)^\top M(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U \mathbf{u}. \tag{10}$$

Derivative vs. U-gradient (Riesz representative). We equip the discrete control space $U_h \cong \mathbb{R}^n$ with the inner product

$$\langle \mathbf{a}, \mathbf{b} \rangle_U := \mathbf{a}^\top M_U \mathbf{b}, \qquad \|\mathbf{a}\|_U := \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle_U}. \tag{11}$$

The (Fréchet) derivative $F'(\mathbf{u})$ is a linear functional on U_h, i.e., $F'(\mathbf{u}) \in U_h^*$. The U-gradient $\nabla_U F(\mathbf{u}) \in U_h$ is defined as the Riesz representative of $F'(\mathbf{u})$ with respect to $\langle \cdot, \cdot \rangle_U$:

$$\langle \nabla_U F(\mathbf{u}), \mathbf{v} \rangle_U = F'(\mathbf{u})[\mathbf{v}] \qquad \forall \mathbf{v} \in \mathbb{R}^n. \tag{12}$$

Connection to ∇F in coefficient coordinates. Sometimes it is convenient to identify the derivative $F'(\mathbf{u})$ with a vector via the standard Euclidean inner product $\langle \mathbf{a}, \mathbf{b} \rangle_2 = \mathbf{a}^\top \mathbf{b}$. We denote this coordinate gradient by $\nabla F(\mathbf{u}) \in \mathbb{R}^n$, defined through

$$F'(\mathbf{u})[\mathbf{v}] = \langle \nabla F(\mathbf{u}), \mathbf{v} \rangle_2 = \nabla F(\mathbf{u})^\top \mathbf{v} \qquad \forall \mathbf{v} \in \mathbb{R}^n. \tag{13}$$

Comparing with the defining identity for $\nabla_U F$ gives the Riesz-map relation

$$\nabla F(\mathbf{u}) = M_U \nabla_U F(\mathbf{u}). \tag{14}$$

Adjoint formula and explicit expression. Using the adjoint variable $\mathbf{p}$ defined by

$$A\mathbf{p} = M(\mathbf{y}(\mathbf{u}) - \mathbf{y}_d), \tag{15}$$

one obtains for all directions $\mathbf{v}$

$$F'(\mathbf{u})[\mathbf{v}] = (B^\top \mathbf{p} + \beta M_U \mathbf{u})^\top \mathbf{v}. \tag{16}$$

Therefore, the U-gradient (i.e., the Riesz representative in the M_U-metric) is

$$\nabla_U F(\mathbf{u}) = M_U^{-1} B^\top \mathbf{p} + \beta \mathbf{u}. \tag{17}$$

The optimality condition is

$$\nabla_U F(\mathbf{u}_*) = \mathbf{0} \quad \Longleftrightarrow \quad B^\top \mathbf{p}_* + \beta M_U \mathbf{u}_* = \mathbf{0}. \tag{18}$$

Quadratic structure and stationarity system. Since $\mathbf{y}(\mathbf{u})$ depends linearly on $\mathbf{u}$, the reduced problem is a strictly convex quadratic program. Indeed, with the control-to-state operator $S := A^{-1}B$ we have

$$F(\mathbf{u}) = \tfrac{1}{2}(S\mathbf{u} - \mathbf{y}_d)^\top M(S\mathbf{u} - \mathbf{y}_d) + \tfrac{\beta}{2}\mathbf{u}^\top M_U \mathbf{u} \tag{19}$$

$$= \tfrac{1}{2}\mathbf{u}^\top (S^\top M S + \beta M_U)\mathbf{u} - \left(S^\top M \mathbf{y}_d\right)^\top \mathbf{u} + \tfrac{1}{2}\mathbf{y}_d^\top M \mathbf{y}_d. \tag{20}$$

Using $S = A^{-1}B$ and symmetry of A, this can be written as the unconstrained quadratic program

$$\min_{\mathbf{u}\in\mathbb{R}^n} \tfrac{1}{2}\mathbf{u}^\top Q\mathbf{u} - \mathbf{b}^\top \mathbf{u} + \text{const}, \qquad Q := B^\top A^{-1} M A^{-1} B + \beta M_U, \quad \mathbf{b} := B^\top A^{-1} M \mathbf{y}_d. \tag{21}$$

Since $B^\top A^{-1} M A^{-1} B$ is symmetric positive semidefinite and βM_U is symmetric positive definite for $\beta > 0$, it follows that Q is symmetric positive definite, hence F is strictly convex. Here the constant term is const $= \frac{1}{2}\mathbf{y}_d^\top M \mathbf{y}_d$, which does not depend on $\mathbf{u}$ and therefore does not influence the minimizer. Eliminating $\mathbf{y}$ and $\mathbf{p}$ from the optimality condition yields the symmetric positive definite (SPD) linear system

$$Q\mathbf{u}_* = \mathbf{b}. \tag{22}$$

Lipschitz constant L and strong convexity modulus m. The constants L and m quantify smoothness and strong convexity of the reduced functional in the control metric. More precisely, L is the Lipschitz constant of the U-gradient,

$$\|\nabla_U F(\mathbf{u}) - \nabla_U F(\mathbf{v})\|_U \le L\|\mathbf{u} - \mathbf{v}\|_U \qquad \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n, \tag{23}$$

and $m > 0$ is the strong convexity modulus,

$$F(\mathbf{v}) \ge F(\mathbf{u}) + \langle \nabla_U F(\mathbf{u}), \mathbf{v} - \mathbf{u}\rangle_U + \tfrac{m}{2}\|\mathbf{v} - \mathbf{u}\|_U^2 \qquad \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n. \tag{24}$$

For the quadratic reduced problem, the U-gradient's U-Hessian is

$$H\mathbf{v} = \nabla_U^2 F(\mathbf{u})\mathbf{v} = M_U^{-1} Q\mathbf{v} = M_U^{-1} B^\top A^{-1} M A^{-1} B\mathbf{v} + \beta\mathbf{v}. \tag{25}$$

This operator is self-adjoint and SPD with respect to $\langle\cdot,\cdot\rangle_U$. Accordingly, m and L are the extreme generalized eigenvalues of the pencil (Q, M_U),

$$Q\mathbf{v} = \lambda M_U \mathbf{v}, \qquad m = \lambda_{\min},\ L = \lambda_{\max}, \tag{26}$$

equivalently the extreme eigenvalues of the SPD matrix $M_U^{-1/2} Q M_U^{-1/2}$. Their ratio $\kappa = L/m$ is the condition number.

4 Optimization Methods

Unless stated otherwise, we initialize at $\mathbf{u}_0 = \nabla_U F(\mathbf{0})$ (as in the code) and stop when $\|\nabla_U F(\mathbf{u}_k)\|_U \le \texttt{tol}_{abs} + \texttt{tol}_{rel}\|\nabla_U F(\mathbf{0})\|_U$ with $\texttt{tol}_{abs} = 10^{-5}$ and $\texttt{tol}_{rel} = 10^{-3}$ or a maximum number of 1000 iterations is reached.

4.1 Steepest Descent/ Gradient method with constant Stepsize $\alpha = 1/L$ (GD)

For an L-Lipschitz gradient in the U-metric, the fixed-step update

$$\mathbf{u}_{k+1} = \mathbf{u}_k - \alpha\,\nabla_U F(\mathbf{u}_k), \quad \alpha = 1/L. \tag{27}$$

Algorithm 1: Steepest Descent with constant stepsize $\alpha = 1/L$

Data: tolerance `tol` > 0, `max_iter`, model with $\nabla_U F$, $\|\cdot\|_U$
Result: sequence $(\mathbf{u}_k)$ converging to a minimizer
$\mathbf{u} \leftarrow \mathbf{u}_0 = \nabla_U F(\mathbf{0})$
Estimate L by power iteration; $\alpha \leftarrow 1/L$;
for $k = 0, 1, \ldots$ **do**
  $\mathbf{g} \leftarrow \nabla_U F(\mathbf{u})$; $g_n \leftarrow \|\mathbf{g}\|_U$;
  if $g_n \leq$ *tol* **then**
    break
  end
  $\mathbf{u} \leftarrow \mathbf{u} - \alpha\, \mathbf{g}$;
end

4.2 Steepest Descent with Barzilai-Borwein stepsize strategy (BB)

BB methods approximate the inverse curvature along the last step by selecting the steplength from secant information in the U-inner product. Define $\mathbf{s}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$, $\mathbf{d}_k = \nabla_U F(\mathbf{u}_k) - \nabla_U F(\mathbf{u}_{k-1})$, and the U-inner product $\langle \cdot, \cdot \rangle_U$. The two classical choices are

$$\alpha_k^{\mathrm{BB1}} = \frac{\langle \mathbf{s}_k, \mathbf{s}_k \rangle_U}{\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U}, \qquad \alpha_k^{\mathrm{BB2}} = \frac{\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U}{\langle \mathbf{d}_k, \mathbf{d}_k \rangle_U}. \tag{28}$$

We alternate BB1/BB2 per iteration and fall back to $\alpha_k = 1$ if $\left|\langle \mathbf{s}_k, \mathbf{d}_k \rangle_U\right|$ is numerically tiny. The update is $\mathbf{u}_{k+1} = \mathbf{u}_k - \alpha_k\, \nabla_U F(\mathbf{u}_k)$.

Algorithm 2: Steepest Descent with Barzilai-Borwein stepsize strategy

Data: `tol` > 0, `max_iter`, model with $\nabla_U F$, $\langle \cdot, \cdot \rangle_U$, $\|\cdot\|_U$
Result: sequence $(\mathbf{u}_k)$
$\mathbf{u}_{-1} \leftarrow \mathbf{0}$; $\mathbf{g}_{-1} \leftarrow \nabla_U F(\mathbf{0})$;
$\mathbf{u}_0 \leftarrow \mathbf{u}_0 = \nabla_U F(\mathbf{0})$;
$\mathbf{g}_0 \leftarrow \nabla_U F(\mathbf{u}_0)$
for $k = 0, 1, \ldots$ **do**
  if $\|\mathbf{g}_k\|_U \leq$ *tol* **then**
    break
  end
  $\mathbf{s}_k \leftarrow \mathbf{u}_k - \mathbf{u}_{k-1}$; $\mathbf{d}_k \leftarrow \mathbf{g}_k - \mathbf{g}_{k-1}$; $sd \leftarrow \langle \mathbf{s}_k, \mathbf{d}_k \rangle_U$;
  if $|sd| < \varepsilon$ **then**
    $\alpha_k \leftarrow 1$;
  else
    if k *is even* **then**
      $\alpha_k \leftarrow sd / \max\{\langle \mathbf{d}_k, \mathbf{d}_k \rangle_U, \varepsilon\}$;
    else
      $\alpha_k \leftarrow \langle \mathbf{s}_k, \mathbf{s}_k \rangle_U / sd$;
    end
  end
  $\mathbf{u}_{k+1} \leftarrow \mathbf{u}_k - \alpha_k\, \mathbf{g}_k$;
  $\mathbf{u}_{k-1} \leftarrow \mathbf{u}_k$; $\mathbf{g}_{k-1} \leftarrow \mathbf{g}_k$;
  $\mathbf{u}_k \leftarrow \mathbf{u}_{k+1}$; $\mathbf{g}_k \leftarrow \nabla_U F(\mathbf{u}_k)$;
end

4.3 Nesterov Acceleration with constant parameters

Set

$$\alpha = 1/L, \qquad \kappa = L/m, \qquad \beta = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}.$$

Then the constant-parameter Nesterov scheme reads

$$\mathbf{y}_k = \mathbf{u}_k + \beta(\mathbf{u}_k - \mathbf{u}_{k-1}), \qquad \mathbf{u}_{k+1} = \mathbf{y}_k - \alpha \nabla_U F(\mathbf{u}_k). \tag{29}$$

Algorithm 3: Nesterov Acceleration with constant parameters

```
Data: tol > 0, max_iter, model with ∇_U F, M_U, Hessian action H
Result: sequence (u_k)
Estimate (L, m) from generalized eigenpairs of (Q, M_U); α ← 1/L; κ ← L/m;
 β ← (√κ − 1)/(√κ + 1)
u_{−1} ← u_0; u ← u_0;
for k = 0, 1, ... do
    g ← ∇_U F(u); g_n ← ‖g‖_U;
    if g_n ≤ tol then
        break
    end
    y ← u + β(u − u_{k−1});
    u_{k+1} ← y − α g;
end
```

4.4 Conjugate Gradient method (CG)

The reduced problem is strictly convex and quadratic. The first-order optimality condition $\nabla_U F(\mathbf{u}_*) = 0$ can be written as an SPD linear system in the control unknowns,

$$Q\mathbf{u}_* = B^\top A^{-1} M \mathbf{y}_d, \qquad Q := B^\top A^{-1} M A^{-1} B + \beta M_U. \tag{30}$$

In the implementation, we solve this system with the Conjugate Gradient method in a matrix-free manner (Hessian action only). Concretely, we apply $Q\mathbf{v}$ via the reduced Hessian action and the control metric, and run the standard CG recurrences while tracking the residual. We use a standard relative residual stopping criterion (as in SciPy's `scipy.sparse.linalg.cg`): with residual $\mathbf{r}_k := \mathbf{b} - Q\mathbf{u}_k$, we stop once

$$\|\mathbf{r}_k\|_2 \le \texttt{rtol}\, \|\mathbf{b}\|_2,$$

and we set $\texttt{rtol} = 10^{-4}$ in all runs. Moreover, since $\nabla_U F(\mathbf{u}) = M_U^{-1}(Q\mathbf{u} - \mathbf{b}) = -M_U^{-1}\mathbf{r}$, the reduced gradient norm can be obtained directly from the residual via

$$\|\nabla_U F(\mathbf{u}_k)\|_U = \sqrt{\mathbf{r}_k^\top M_U^{-1} \mathbf{r}_k},$$

which we record as the convergence curve in Figure 2.

5 Numerical Results

We report results on a representative external mesh with size $h = 0.02$ (here $n = 3093$ control degrees of freedom). We use a Tikhonov parameter $\beta = 10^{-3}$. The first-order methods (BB, GD, Nesterov) are initialized at $\mathbf{u}_0 = \nabla_U F(\mathbf{0})$ and stopped when $\|\nabla_U F(\mathbf{u}_k)\|_U \le \texttt{tol}_{abs} + \texttt{tol}_{rel}\|\nabla_U F(\mathbf{0})\|_U$ with $\texttt{tol}_{abs} = 10^{-5}$ and $\texttt{tol}_{rel} = 10^{-3}$. In addition, we solve the SPD stationarity system $Q\mathbf{u} = \mathbf{b}$ with CG.

Algorithm 4: CG for $Q\mathbf{u} = \mathbf{b}$ (matrix-free)

Data: model with A^{-1} solves, matrices B, M, M_U, Hessian action H; `rtol` > 0
Result: $\mathbf{u}_*$
$\mathbf{b} \leftarrow B^\top A^{-1}(M\mathbf{y}_d)$;
Define operator $Q\mathbf{v} := M_U(H\mathbf{v})$ (equivalently $B^\top A^{-1} M A^{-1} B\mathbf{v} + \beta M_U \mathbf{v}$);
Call CG to solve $Q\mathbf{u} = \mathbf{b}$ to relative tolerance `rtol` $= 10^{-4}$;

Calculation and results for m and L. The smoothness/strong-convexity constants are defined as the extreme generalized eigenvalues of (Q, M_U), $Q\mathbf{v} = \lambda M_U \mathbf{v}$, i.e. $L = \lambda_{\max}$ and $m = \lambda_{\min}$. In the implementation, both are estimated with `scipy.sparse.linalg.eigsh` applied to the generalized eigenproblem using a matrix-free Q-operator. On the representative mesh $h = 0.02$ we obtain

$$L \approx 1.31636 \times 10^{-3}, \qquad m \approx 1.0 \times 10^{-3}, \qquad \kappa = L/m \approx 1.31636.$$

As expected for $Q = B^\top A^{-1} M A^{-1} B + \beta M_U$ with $\beta > 0$, the smallest generalized eigenvalue is close to β.

Final cost and iteration table. Table 1 compares final costs and iteration counts for the representative mesh. All methods reach essentially the same optimal value; differences are at the level of numerical tolerances.

Method	Final cost $F(\mathbf{u}_k)$	Iterations
CG ($Q\mathbf{u} = \mathbf{b}$)	0.0161709	6
BB	0.0161709	1
GD ($1/L$)	0.0161709	1
Nesterov	0.0161709	1

Table 1: Representative-mesh results ($h = 0.02$). All methods reach the same optimum up to numerical precision; GD, BB, and Nesterov attain the stopping criterion in only one iteration.

Gradient norm convergence plot. Figure 2 shows the convergence histories in terms of the U-norm of the reduced gradient, $\|\nabla_U F(\mathbf{u}_k)\|_U$, which is the stopping quantity for BB/GD/Nesterov. For CG, we reconstruct the same quantity from the stationarity-system residual via $\|\nabla_U F(\mathbf{u}_k)\|_U = \sqrt{\mathbf{r}_k^\top M_U^{-1} \mathbf{r}_k}$; this additional M_U^{-1}-solve per CG iteration is computed only to enable a direct convergence-curve comparison with the first-order methods.

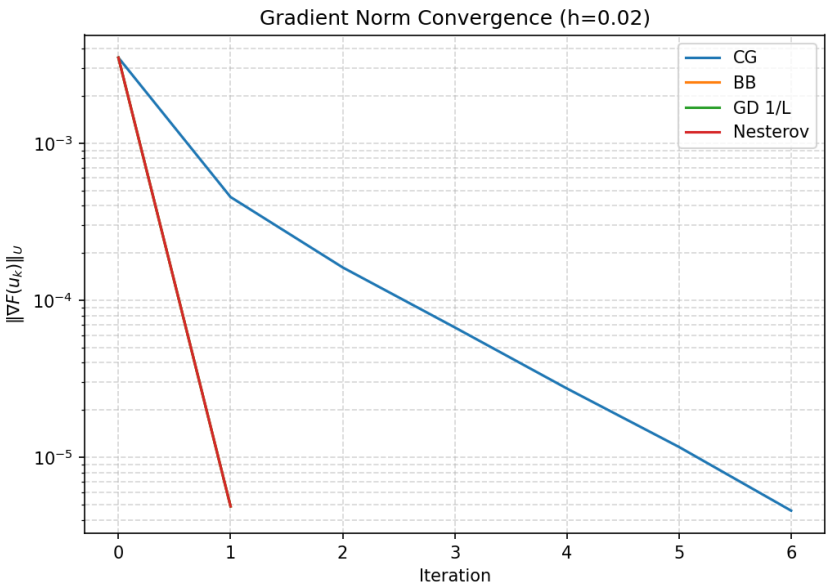


Figure 2: Convergence of gradient norm for a representative mesh ($h = 0.02$). The control-space norm $\|\cdot\|_U$ is induced by M_U. The GD, BB, and Nesterov curves are nearly identical and may visually overlap.

Time measurement. We measure wall-clock time with `time.perf_counter`: for GD and Nesterov we separate the time spent in spectral estimates (t_L and t_m) from the subsequent iteration runtime (t_{iter}). Table 2 reports these components and $t_{total} = t_L + t_m + t_{iter}$. For CG, we report the iteration time for solving $Q\mathbf{u} = \mathbf{b}$ and explicitly *exclude* the additional work needed to record the gradient-norm history (one M_U^{-1}-solve per iteration to evaluate $\sqrt{\mathbf{r}_k^\top M_U^{-1}\mathbf{r}_k}$), since this diagnostic is used only for the convergence-plot comparison.

Method	t_L [s]	t_m [s]	t_{iter} [s]	t_{total} [s]
CG	0.0000	0.0000	0.0152	0.0152
BB	0.0000	0.0000	0.0082	0.0082
GD	0.0596	0.0000	0.0053	0.0649
Nesterov	0.0596	0.0652	0.0052	0.1300

Table 2: Timing breakdown on the representative mesh $h = 0.02$. Here t_L and t_m denote time spent estimating spectral constants (if used), t_{iter} is the optimization runtime excluding estimates, and $t_{total} = t_L + t_m + t_{iter}$.

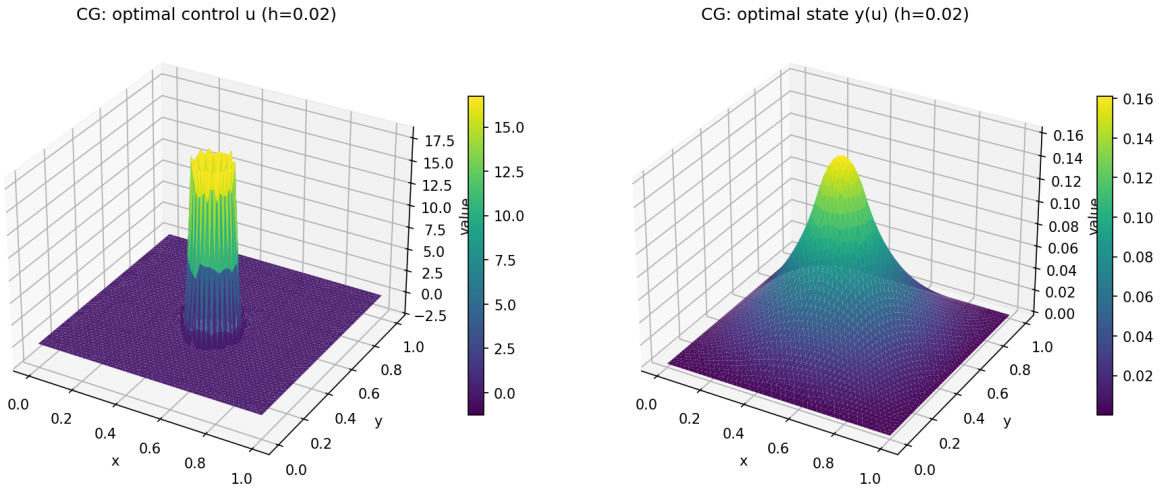


Figure 3: Optimized control u_* (left) and the corresponding state $y(u_*)$ (right) computed by CG on the stationarity system, shown for a representative mesh ($h = 0.02$).

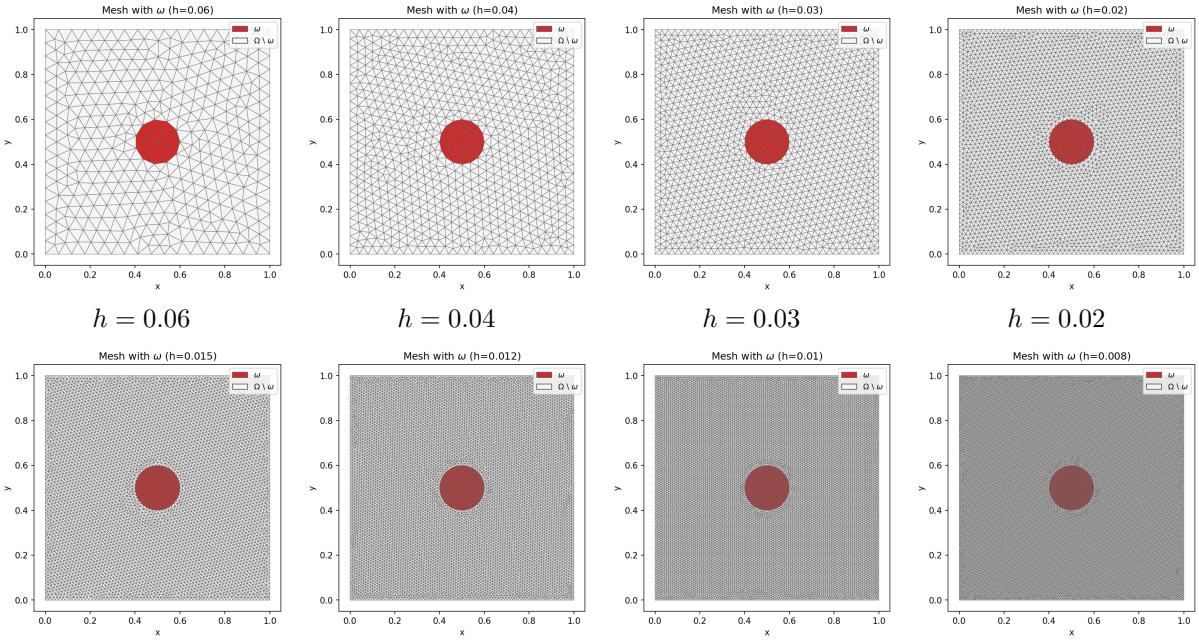
Interpretation. Table 1 shows that all four methods reach essentially the same minimizer (up to numerical precision), as expected for a strictly convex quadratic problem with a unique solution. Figure 2 confirms rapid decay of the first-order stopping quantity $\|\nabla_U F(\mathbf{u}_k)\|_U$ for GD, BB, and Nesterov; on the representative mesh, the three curves are nearly indistinguishable.

Table 2 highlights the runtime trade-off between iteration count and one-time preprocessing. Fixed-step GD requires an estimate of L, and Nesterov requires estimates of both m and L, so their totals include t_L and t_m. BB and CG avoid this preprocessing. On the representative mesh, BB achieves the smallest wall-clock time because it satisfies the stopping criterion in one iteration without spectral precomputation.

Figure 3 provides a qualitative view of the resulting optimal control and associated state.

6 Mesh Independence

We study how the discretized solution and the observed optimizer performance depend on mesh resolution. We use a fixed family of *pre-generated* triangular meshes of $\Omega = [0, 1]^2$ with a circular control region ω for $h \in \{0.06, 0.04, 0.03, 0.02, 0.015, 0.012, 0.01, 0.008\}$; the meshes are visualized in Figure 4.



$h = 0.06$ $h = 0.04$ $h = 0.03$ $h = 0.02$

$h = 0.015$ $h = 0.012$ $h = 0.01$ $h = 0.008$

Figure 4: Meshes used in the mesh-independence study.

Table 3 summarizes objective values and solution norms across meshes. Here $u_*^{(h)}$ denotes the discrete minimizer for mesh size h, approximated by solving the stationarity system with CG. Under refinement, $F(u_*^{(h)})$, $\|u_*^{(h)}\|_U$, and $\|y(u_*^{(h)})\|_{L^2}$ vary only mildly, indicating proximity to a mesh-independent limit in this experiment. Figure 5 shows the same trend graphically and indicates that all methods attain essentially the same final objective on each mesh (up to stopping tolerances).

h	DOFs	$F(u_*^{(h)})$	$\|u_*^{(h)}\|_U$	$\|y(u_*^{(h)})\|_{L^2}$
0.06	402	0.0165171	2.549350	0.042344
0.04	820	0.0163319	2.607212	0.044949
0.03	1434	0.0162550	2.632510	0.046133
0.02	3093	0.0161709	2.654889	0.047219
0.015	5392	0.0161253	2.665354	0.047746
0.012	8438	0.0161035	2.671025	0.048028
0.01	11899	0.0160847	2.675265	0.048244
0.008	18497	0.0160695	2.678685	0.048419

Table 3: Per-mesh summary of objective and norms. DOFs denotes the number of P1 degrees of freedom.

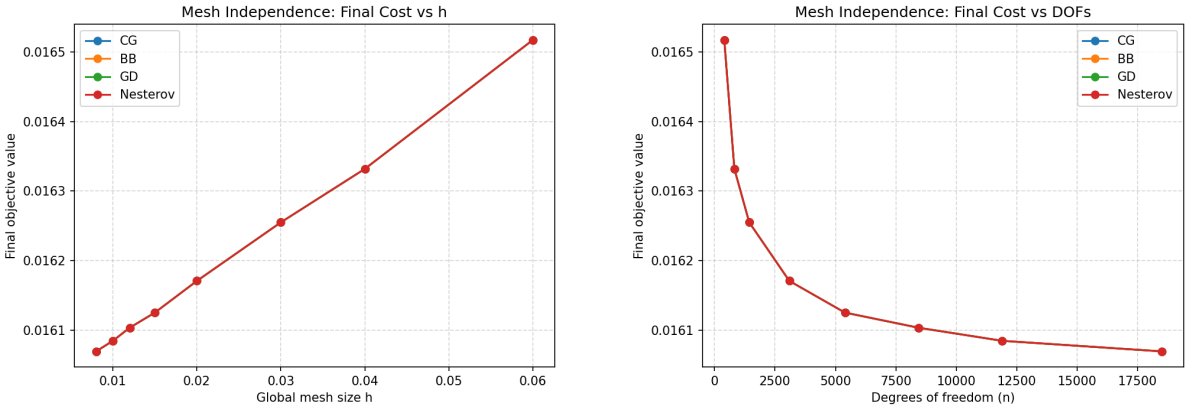


Figure 5: Mesh independence study: final objective value versus mesh size h (left) and versus degrees of freedom (right), for all methods (CG, BB, GD, Nesterov).

Table 4 and Figure 6 summarize iteration counts across meshes. For BB/GD/Nesterov, counts refer to update steps. These counts are not directly comparable to CG counts because the stopping rules differ: first-order methods use $\|\nabla_U F(\mathbf{u}_k)\|_U \leq \texttt{tol}$, whereas CG uses a relative residual criterion for $Q\mathbf{u} = \mathbf{b}$. Over the tested mesh range, first-order counts remain fairly stable, while CG counts vary moderately with h.

h	CG	BB	GD	Nesterov
0.06	7	1	1	1
0.04	8	1	1	1
0.03	7	1	1	1
0.02	6	1	1	1
0.015	6	1	1	1
0.012	6	1	1	1
0.01	6	1	1	1
0.008	6	1	1	1

Table 4: Iterations per mesh: CG iterations for the stationarity solve and iteration counts for the first-order methods.

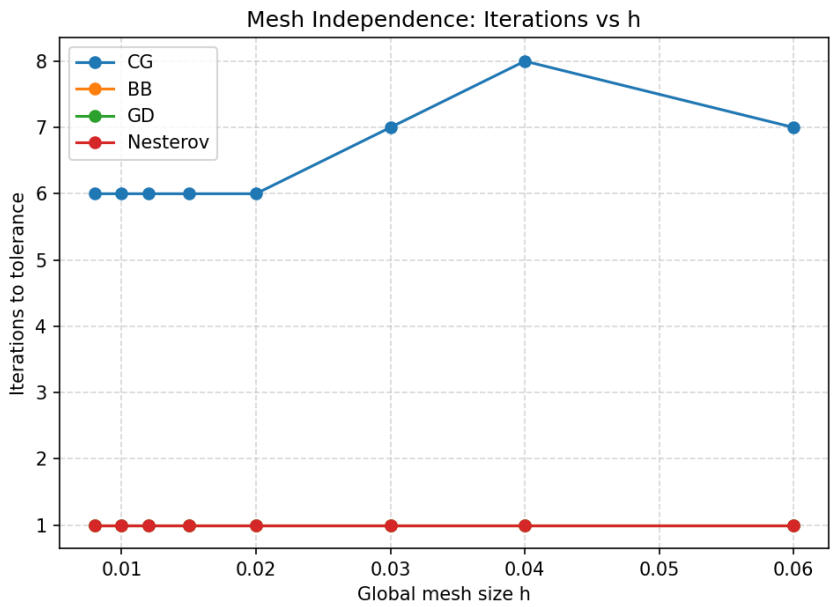


Figure 6: Mesh independence study: iteration counts versus mesh size h (CG iterations for the stationarity solve and first-order iterations for BB/GD/Nesterov).

In addition to iteration counts, we compare wall-clock times to reach the respective stopping criteria. Figure 7 reports total time (including one-time spectral estimates where applicable). Figure 8 isolates the estimate time $t_L + t_m$, and Figure 9 reports the iteration time excluding these estimate costs.

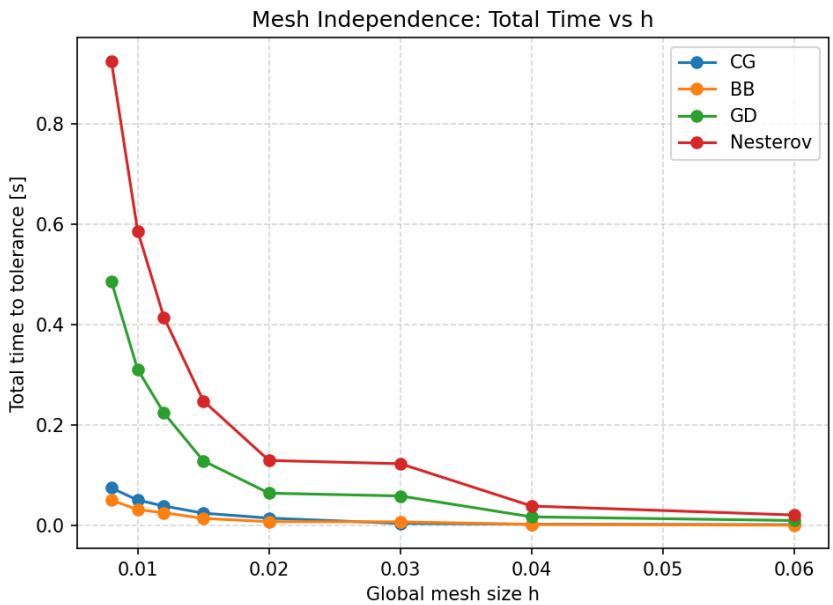


Figure 7: Mesh independence study: total time to tolerance versus mesh size h (including estimate time where applicable).

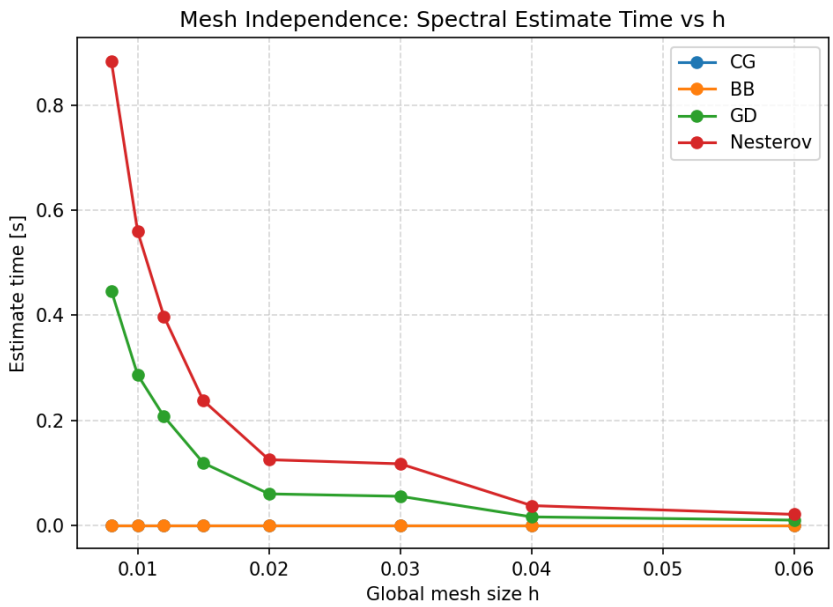


Figure 8: Mesh independence study: time spent estimating spectral constants (here reported as $t_L + t_m$) versus mesh size h.

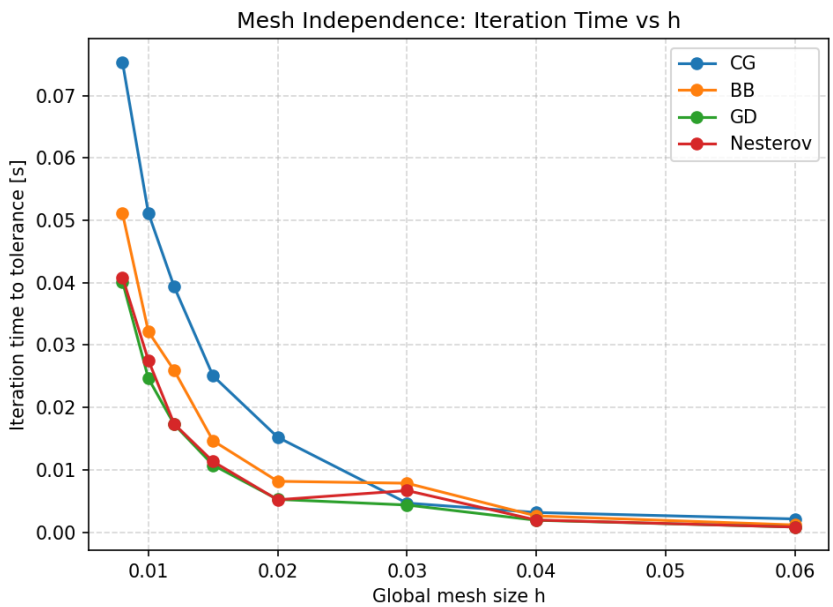
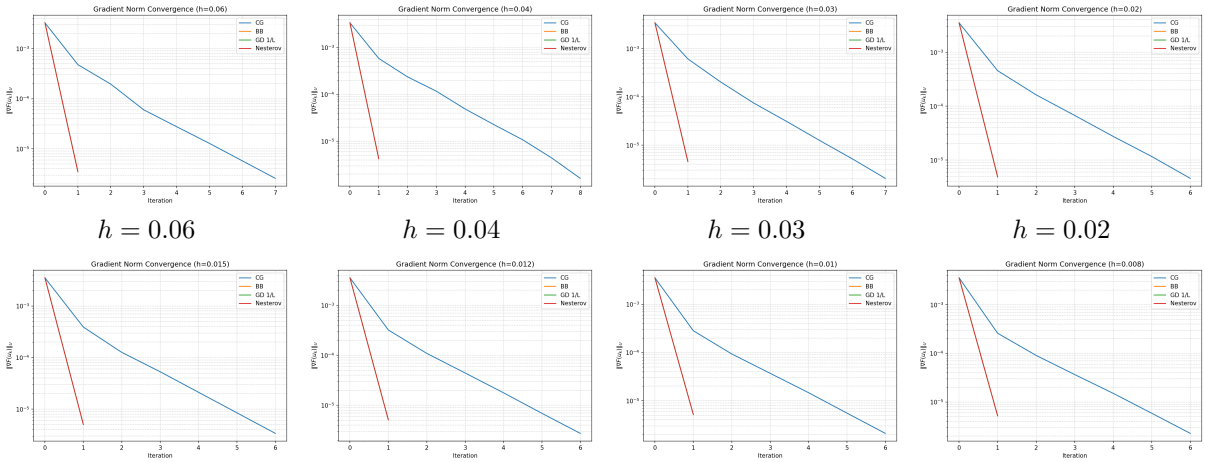


Figure 9: Mesh independence study: iteration time to tolerance versus mesh size h (excluding spectral estimate time t_L and t_m where applicable).

Finally, Figure 10 shows the per-mesh convergence curves $\|\nabla_U F(\mathbf{u}_k)\|_U$, illustrating that the qualitative convergence behavior remains similar across the mesh family.



$h = 0.06$ $h = 0.04$ $h = 0.03$ $h = 0.02$

$h = 0.015$ $h = 0.012$ $h = 0.01$ $h = 0.008$

Figure 10: Gradient norm convergence $\|\nabla_U F(\mathbf{u}_k)\|_U$ for each mesh.